

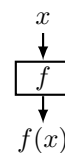
L1.1a Functions & Catalog (§1.1–1.2)

What you need to know

A FUNCTION, FOUR WAYS

Algebraic · numerical · graphical · verbal — one object, four views. f gives each x in D **exactly one** $f(x)$ in E ; its graph is the set

$$\{ (x, f(x)) \mid x \in D \}$$



“Think of f as a machine — the output must be defined.”

D = every input R = every output **VLT**: no vertical line hits it twice.

DOMAIN & RANGE

even root's inside ≥ 0 denominator $\neq 0$

D off the formula, R off the outputs. **Both notations:** $\{x \mid x \neq 0, 1\} = (-\infty, 0) \cup (0, 1) \cup (1, \infty)$
Sign probes, not a sign chart: sign each factor @-1, @0.5, @10, then read the whole.

“Intersection cuts out just the overlap of the two domain intervals. Union gives you any place that either of them goes.”

The catalog — nine families

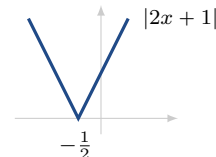
Family	Form	Domain / range	What marks it
Linear	$mx + b$	$D = R = (-\infty, \infty)$	$m = \frac{\Delta y}{\Delta x}$, constant everywhere
Polynomial	$a_n x^n + \dots + a_0$	$D = (-\infty, \infty)$	smooth, $\leq n - 1$ turns; factor with Loh
Power	x^n	even n : $R = [0, \infty)$; odd n : $R = (-\infty, \infty)$	x^6 is a <i>steeper</i> x^2 , not a new shape
Root	$\sqrt[n]{x}$	even n : $D = [0, \infty)$; odd n : $D = (-\infty, \infty)$	odd roots have negative domains
Rational	$\frac{P(x)}{Q(x)}$	D : every x with $Q(x) \neq 0$	Z@ · VA@ · HE@ · HA
Algebraic	$\sqrt{x^2 + 1}$	whatever survives every root and denominator	polynomials put through + - × / and roots
Trig	$\sin x, \cos x, \tan x$	$\sin, \cos$: $R = [-1, 1]$; $\tan$: $x \neq \frac{\pi}{2} + \pi k$	$\sin, \tan$ odd; $\cos$ even; all periodic
Exponential	$b^x, b > 0$	$D = (-\infty, \infty), R = (0, \infty)$	HA y=0 ; the variable is the exponent
Logarithmic	$\log_b x$	$D = (0, \infty), R = (-\infty, \infty)$	VA@@, Z@1 ; “what exponent?”

Loh, never the quadratic formula: divide by the leading coefficient, then $m = -\frac{b}{a}$, $d = \sqrt{m^2 - c}$, roots $m \pm d$.
 E.g. $3x^2 + 2x - 1 \rightarrow x^2 + \frac{2}{3}x - \frac{1}{3}$: $m = -\frac{1}{3}$, $d = \frac{2}{3}$, roots $\frac{1}{3}, -1$, so $(3x - 1)(x + 1)$.

HA, three cases: $\deg(\text{bot}) > \deg(\text{top}) \Rightarrow \text{HA y=0}$ · equal $\Rightarrow$ ratio of leading coefficients · $\deg(\text{bot}) < \deg(\text{top}) \Rightarrow$ long division, **SA**.

PIECEWISE & ABSOLUTE VALUE

$$|x| = \begin{cases} x & x \geq 0 \\ -x & x < 0 \end{cases}$$



“chop and reflect at the vertex”

Four substitutions. $2x + 1$ goes in for *each* x — both formulas **and** both restrictions, which are inequalities to solve.

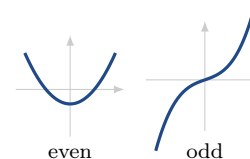
$$|2x + 1| = \begin{cases} 2x + 1 & 2x + 1 \geq 0 \Rightarrow x \geq -\frac{1}{2} \\ -(2x + 1) & 2x + 1 < 0 \Rightarrow x < -\frac{1}{2} \end{cases}$$

SYMMETRY

$$f(-x) = f(x) \text{ even} \quad f(-x) = -f(x) \text{ odd}$$

“symmetric = even, antisymmetric = odd”

Test the algebra, not the picture. Most functions are **neither** — $x - x^2$.

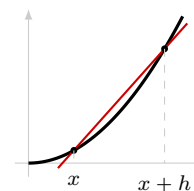


THE DIFFERENCE QUOTIENT

$$m = \frac{\Delta y}{\Delta x} = \frac{f(x + h) - f(x)}{h}$$

“average rate of change — the slope of the secant”

$x^2 + x \Rightarrow 2x + 1 + h$. A line keeps neither h nor x :
 $4x - 1 \Rightarrow 4$.



WATCH OUT

- ✗ $\sqrt[3]{-8}$ is undefined → odd roots have negative domains — $\sqrt[3]{-8} = -2$
- ✗ $|2x + 1|$ takes two substitutions → four — both formulas *and* both restrictions
- ✗ a cancelled factor comes back → that input is still gone — it is a hole, HEO, not a VA
- ✗ every function is even or odd → most are neither — $x - x^2$
- ✗ the difference quotient is the tangent slope → it is the *secant* slope — an average over a step h
- ✗ $\cup$ and $\cap$ both mean “combine” → $\cap$ keeps only the overlap; $\cup$ keeps anywhere either goes
- ✗ use the quadratic formula → Loh, always